

# Ratios, Proportion & Percentages



## Simplifying ratios

- 1 Simplify the ratio 18 : 24.
  - 2 Simplify the ratio 45 : 60 : 75.
  - 3 Simplify the ratio 36 : 8.
  - 4 Express "12 boys to 18 girls" as a ratio in simplest form.
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## Sharing a quantity in a given ratio

- 5 Share 150 in the ratio 2 : 3.
  - 6 Share 220 among three people in the ratio 3 : 4 : 4.
  - 7 A prize of 900 riyals is shared between Sara and Layla in the ratio 5 : 4. Find each share.
  - 8 Divide 84 kg of flour in the ratio 1 : 2 : 4.
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## Equivalent ratios and proportions

- 9 If  $3 : 5 = x : 20$ , find  $x$ .
  - 10 If  $a : 7 = 12 : 21$ , find  $a$ .
  - 11 A map has scale 1 : 50,000. A distance of 3 cm on the map represents how many meters in reality?
  - 12 Solve for  $y$ :  $\frac{4}{9} = \frac{y}{63}$ .
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## Percent of a number

- 13 Find 35% of 240.
- 14 Find 12% of 150.
- 15 What is 8% of 650?

- 16 Find 150% of 40.
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### Converting between fractions, decimals and percentages

- 17 Write  $\frac{3}{8}$  as a percent.
- 18 Write 0.45 as a fraction in lowest terms.
- 19 Write 62.5% as a fraction in lowest terms.
- 20 Write  $\frac{7}{40}$  as a decimal.
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### Percentage increase and decrease

- 21 Increase 80 by 25%.
- 22 Decrease 150 by 20%.
- 23 A salary of 4,000 riyals increases by 6%. Find the new salary.
- 24 A population of 2,500 decreases by 12%. Find the new population.
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### Finding percentage change and original values

- 25 A price rises from 80 to 92. Find the percentage increase.
- 26 A quantity falls from 250 to 200. Find the percentage decrease.
- 27 After a 20% discount, a jacket costs 96 riyals. Find the original price.
- 28 After a 15% increase, a value becomes 138. Find the original value.
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### Successive percentage changes

- 29 A price of 100 rises 10%, then falls 10%. Find the final price.
- 30 A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.
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- 31 A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.
- 32 A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.
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### Direct proportion

- 33  $y$  is directly proportional to  $x$ , and  $y = 15$  when  $x = 3$ . Find  $y$  when  $x = 8$ .
- 34 4 kg of apples cost 28 riyals. Find the cost of 7 kg at the same rate.
- 35 A car uses 8 liters of fuel to travel 96 km. How far can it travel on 20 liters at the same rate?
- 36 6 identical pens cost 27 riyals. Find the cost of 10 pens.
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### Inverse proportion

- 37 4 workers can complete a job in 15 days. How many days would 6 workers take, working at the same rate?
- 38 A car covers a fixed distance in 5 hours at 60 km/h. How long would the same trip take at 75 km/h?
- 39 8 taps fill a pool in 12 hours. How long would 3 taps take, running at the same rate?
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### Best value and unit price comparisons

- 40 A 3 kg bag of rice costs 27 riyals, and a 5 kg bag costs 42.50 riyals. Which is the better value per kilogram?
- 41 Store A sells 6 pens for 15 riyals; Store B sells 8 pens for 19.20 riyals. Which store offers the lower price per pen?
- 42 A 2-liter bottle of juice costs 14 riyals. Find the price per 100 mL.
- 43 Which is the better deal: 12 eggs for 9 riyals, or 18 eggs for 12.60 riyals?
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## Word problems

- 44 A recipe requires flour and sugar in the ratio 5 : 2. If 350 g of flour is used, how much sugar is needed?
- 45 A shop raises the price of a shirt from 60 to 75 riyals, then during a sale reduces the new price by 20%. Find the sale price.
- 46 In a class of 40 students, 60% are girls. Then 4 more girls join the class. What percent of the class is now girls, to the nearest percent?
- 47 A dealership reduces the price of a car by 15% to 25,500 riyals. Find the original price.
- 48 Ahmed invests 8,000 riyals in an account that grows by 5% each year. Find its value after 2 years.
- 49 A map has scale 1 : 25,000. Two towns are 8.4 cm apart on the map. Find the actual distance between them in kilometers.